

Pergunta 1

Respondida

Pontuou 2,00 de 2,00

Destacar pergunta

What was the first (recorded) software bug and how was it identified? Please justify your answer.

- a) On the 50s, using NATO's advanced bug identification systems
- b) On the 80s, using critical service analysis
- c) On the 40s, manually
- d) On the 90s, using automated testing.

Option C

Back in the 40s, computers filled large rooms and their warmth attracted bugs. These creatures then shorted circuits and caused the computer to malfunction, which lead, on September 9 1945, U.S. Navy officer Grace Hopper to manually find a moth between the relays on the Harvard Mark II computer she was working on.

Pergunta 2

Respondida

Pontuou 2,00 de 2,00

Destacar pergunta

Which one of the following is most adequate to have a software prototype (please justify your answer):

- a) A customer-driven app for ordering food;
- b) A business-to-business software for supporting a supply chain;
- c) A healthcare software to handle critical data;
- d) All of the above

Option D

All of the above options should include a prototyping activities, as this is a crucial step to help elicit requirements for later increments. Early increments of the process should be used to better study the problem in question, by establishing a Prototyping plan, outlining its definition, executing and then evaluating on the result.

Pergunta 3

Respondida

Pontuou 2,00 de 2,00

Retirar destaque

A use case model is (please justify):

- a) The set of use case diagrams plus the sequence diagrams
- b) The set of use case diagrams plus the domain diagrams
- c) The set of use case diagrams only
- d) The set of use case diagrams plus the necessary support documentation

Option D

Use case models are composed by use case diagram(s) plus the associated documentation. A use case model is used to organize functional requirements, in a way closer to the structure of the final system. This is done through the use of use case diagrams that show the system purpose and usefulness, and the corresponding documentation that elaborates and details the information of the use cases.

Pergunta 4

Respondida

Pontuou 2,00 de 2,00

Destacar pergunta

Software Engineering has been a research subject for the past decades. Focusing on software architectures, engineers have learned much about how to build good software. What technique/approach is typically adopted to share such practical knowledge between engineers? Describe why is it a good technique/approach.

- a) Object Oriented Programming.
- b) Reusable Open Source libraries.
- c) Design Patterns.
- d) Sequence Diagrams.

Option C

Design Patterns specify template solutions for common types of problems, describing how modules can/should be organized in order to maintain a good codebase. These are solutions that have worked well before and adapt smoothly and incrementally to new situations. They should be used instead of architecting solutions from scratch, as this is hard and, most times, not effective.

Pergunta 5

Respondida

Pontuou 2,00 de 2,00

Retirar destaque

Agile methodologies are recommended for teams smaller than 10 elements. From the options below, choose which approach should be adopted if there is the need to increase the size of an agile development team and justify your choice.

- a) Adopt a more traditional methodology that was though for a larger team;
- b) Add the additional team members to the team and monitor the results, agile methodologies are able to embrace change;
- c) Development teams should never be changed during the development of a project;
- d) Divide the team into smaller agile teams that handle specific components of the project.

Option D

Agile methodologies are meant to be able to easily adapt to changes, and increasing the size of the development team shouldn't be a problem. Since the previous elements of the developing team already knew eachother dividing them will not be a problem, we just need to introduce the new members to the team and evaluate how the team performs. With more people, but each team with specific components of the project, all teams can keep an agile method and the company would have more people working and making the project better.

Pergunta 6

Respondida

Pontuou 2,00 de 2,00

Destacar pergunta

Historically, software development was an inefficient craft. eXtreme Programming proposed to change that by introducing novel development practices. What option has practices intrinsic to XP's nature? Briefly describe the identified practices.

- a) Mockups, pair programming, continuous integration;
- b) Adopt better programming languages, small teams, design patterns;
- c) Pair programming, design improvement, test first;
- d) Coding standards, documentation with class diagram.

Option C

Pair programming is the activity where two people are programming in the same computer: when one actually writes the code, the other observes the programmer progress and points out anything that might seem unusual, giving feedback and helping him. From time to time the two switch up.

Design improvement (where refactoring is included) is a process in which the a program's source code is restructured without changing the program's functionality.

Test first (test-driven development) is a way of development where the unit tests and written first and only then the necessary code to pass those is developed.

Pergunta 7

Respondida

Pontuou 2,00 de 2,00

Destacar pergunta

Legacy systems are older software systems, developed using obsolete software and hardware technologies, that remain useful for a business. It is often cheaper and less risky to maintain a legacy system than to develop a replacement system using modern technology. What do you think is the main reason for this?

- a) Modern technology is unable to cope with old requirements.
- b) Hiring a new team, proficient in the modern technologies, is too expensive.
- c) The business value of the legacy system still outweighs the technology value.
- d) The maintenance team has too much implicit knowledge on the system.

Option C

Replacing a legacy system with a system using modern technology involves lots of process which are time and money wasting. To verify if the change is worth the business value of the legacy system and the quality of the application should be assessed to help decide if a system should be replaced, transformed or maintained. When the business value still outweighs the technology value the change is not worth making.

Pergunta 8

Respondida

Pontuou 0,00 de 2,00

Destacar pergunta

Software project management is concerned with activities to ensure that software is delivered on time, on budget, and in accordance with the requirements of the organisations developing and procuring the software. One of the activities of project management is risk management: identification, analysis, planning, and monitoring of risks.

Which of the following do you consider the best example of a risk in a software project?

- a) team elements are not self-organized and need a strong leadership
- b) the computers of the developers take a lot of time to do a full build of the software system
- c) the servers infrastructure is outsourced to a private company
- d) all team elements plan to spend one full week together in another continent, 12 hours flight away

Option C

If the servers infrastructure is outsourced to a private company the team won't have control over that infrastructure, which means if any problem occurs they won't be able to do anything about it since it is dependent on another company, which can lead to an over-time delivery. We should try to always have control on the tools we need to develop the projects. Outsourcing to another company can also imply more costs than to hire a team capable of handling that work and the respective hardware needed.

Pergunta 9

Respondida

Pontuou 2,00 de 2,00

🚩 Destacar pergunta

Briefly describe the most important phases and respective activities of RUP that you have used in your app project, explaining their purpose in the project and the corresponding results.

RUP consists of 4 phases: Inception, Elaboration, Construction and Transition. In our project, in the Inception phase we established what was intended to be in the product and what is not. In the elaboration phase we would make and refine all the documentation needed for the project, and create plans for the construction phase so that we didn't need to worry about anything other than coding and getting the features implemented. In the construction phase the coding was done, both the needed for the features and the needed for the corresponding tests. Finally in the transition phase we merged all the branches and created a product release that was ready for the client to test (in this course it was the teacher) through an .apk. We showed the teacher the application and all the implemented features in the sprint, with all the acceptance tests passing. With his feedback we would then tune the product and make it better in the next release.

Pergunta 10

Respondida

Pontuou 1,50 de 2,00

🚩 Destacar pergunta

Briefly describe the most important agile practices that you have used in your app project, explaining their purpose in the project and the corresponding results.

In agile methods the planning is incremental and where the team can adapt quickly to the client requirements (in our case it was the teacher feedback). In the start of each sprint the team would discuss which features to implement, based on the teacher feedback, and the features effort. Extreme Programming (an agile practice) was used a lot through all the project development, mainly Pair Programmig, Refactoring and Collective Code Ownership. Each feature would be assign to at least 2 team members, then those members would try to work on the same day and, since we were mainly at home due to the pandemic, the Pair Programming was made using Discord screen share. With this while one was programming the other(s) would be giving feedback and improving some documentation. This also contributed to the Collective Code Ownership since multiple people were assigned to the same task and would work at the same time, meaning all understood the code and where on par with it. When it comes to Refactoring we always made sure to clean the duplicated code and try to simplify it as much as we could before merging the branches to the main one where releases were made.

All these practices lead to a simpler and easier to understand code, and to a project where everyone was a envolved a lot and always knowing what was going on. Team communication and the teacher feedback were key to our success.